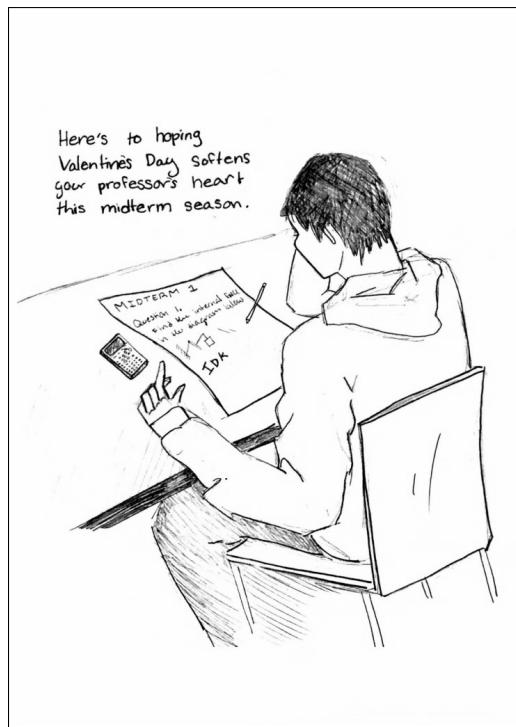


MTH310/MTH520: Graph Theory (GT)

Quiz 2

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Originally created by Lena Siskind, Author at The Michigan Daily

Solve any 2 out of 3 questions in your answer booklet. In case all are attempted, the answer with the minimum marks will be dropped in the final score.

1. Let X and Y be disjoint sets of vertices in a k -connected graph G . Let $f : V(G) \rightarrow \mathbb{N}$ ($0 \in \mathbb{N}$ in this case) be a function such that $\sum_{x \in X} f(x) = \sum_{y \in Y} f(y) = k$. Prove that G has k pairwise independent X - Y paths so that $f(x)$ of them start at x and $f(y)$ of them end at y for each $x \in X$ and $y \in Y$.¹

Hints:

- a) You have seen this type of question before in the tutorials and classes, where you are given a graph G with some extra information, you construct an auxiliary

¹We say that a set of $X - Y$ paths are *pairwise independent* if the only vertices common to a pair of paths are their endpoints.

graph G' based on that information and apply a famous theorem on G' to reach our desired conclusion.

- b) On constructing the auxiliary graph: Suppose you make a new graph with $\lambda \geq 1$ copies of some $v \in V$, say $v_1, \dots, v_\lambda$, and connect each v_i to every $u \in N(v)$ in the original graph. All other vertices and edges remain the same in the new graph. Then, observe that for any $A \subseteq V$, a set of disjoint $\{v_i : 1 \leq i \leq \lambda\} - A$ paths in the new graph, constitutes a set of $v - A$ pairwise independent paths in the original graph.
- c) Also, note that there may be points $x \in X$ and $y \in Y$ such that $f(x) = 0$ or $f(y) = 0$. However, you should be able to argue that without the loss of generality, it is enough to prove the result for the case where for every $x \in X$ and every $y \in Y$, $f(x) > 0$ and $f(y) > 0$.

[5 points]

2. Given a graph G with maximum degree Δ and nonnegative integers d_1, d_2 with $d_1 + d_2 \geq \Delta - 1$. In this question, we will prove that the set V of the vertices of G can be partitioned into sets V_1, V_2 so that $\Delta(G[V_i]) \leq d_i$ for $i \in \{1, 2\}$.

- a) Consider a function w , where $w(V_1, V_2) = \frac{|E(G[V_1])|}{d_1} + \frac{|E(G[V_2])|}{d_2} \forall V_1 \cap V_2 = \phi, V_1 \cup V_2 = V(G)$. Let $(X, Y) \in \arg \min_{\substack{V_1 \cup V_2 = V(G) \\ V_1 \cap V_2 = \phi}} w(V_1, V_2)$.

Assume that $\exists x \in X$ with $d_{G[X]}(x) > d_1$. Based on this vertex x , define $E(x, Y) := |\{\{x, y\} \in E(G) \mid y \in Y\}|$, and show that $\frac{d_{G[X]}(x)}{d_1} > \frac{E(x, Y)}{d_2}$.

- b) Construct another partition (X', Y') such that $w(X', Y') < w(X, Y)$, hence creating our desired contradiction. Show how this completes the proof.

(Hint: Can you exchange something from X to Y to ensure this condition holds?)

[2+3=5 points]

3. In the previous quiz, Q2 (question on hypercube graph Q_n) was considered "harder" and unpopular since only 1 in 7 students attempted it. In order to give a head-start to those students, we will give a direct extension of that question here:

A graph G is called a *median graph* if G is connected and $\forall a, b, c \in V(G)$, $\exists! v \in V(G)$ such that $d_G(a, v) + d_G(v, b) = d_G(a, b)$, $d_G(b, v) + d_G(v, c) = d_G(b, c)$ and $d_G(c, v) + d_G(v, a) = d_G(c, a)$ (v is called the *median* of the vertices a, b, c , and we denote it by $m(a, b, c)$). Prove that Q_n is a median graph $\forall n \in \mathbb{N}$.

Note: $Q_n = (\{0, 1\}^n, \{\{u, v\} \mid u, v \in \{0, 1\}^n \text{ that differ by exactly one coordinate}\})$
(Hint: Remember that $d_{Q_n}(u, v)$ is the number of bit flips between n -bit u and v . Now, for arbitrary $x, y, z \in \{0, 1\}$, define $\text{Maj}(x, y, z)$ to be the symbol 0 or 1 in which the majority of x, y, z take value. Now, given three vertices $\{u, v, w\} \subseteq V(Q_n)$, show that the vertex z such that $z(i) = \text{Maj}(u(i), v(i), w(i))$, for all $1 \leq i \leq n$ satisfies $m(u, v, w) = z$. Do not forget to prove the uniqueness of z as the median.)

[5 points]